Module 03: Lab 03

Exercises

1) After working through all of the code in the lab, run ‘head(training_features)’ and ‘head(test_features)’. Copy and paste the output.

```{r}
head(training_features)
```

```
            [,1]       [,2]       [,3]        [,4]       [,5]       [,6]      [,7]        [,8]       [,9]
[1,] -1.6250271 -0.5496140 -0.9694237 -0.40839628 -0.4972050 -1.0573835 0.6995348  0.29979834 -0.2788510
[2,] -0.2887126 -0.6036596 -0.2887592 -1.12134458 -0.7553537  1.3143730 0.9289750 -0.10713309 -0.7490750
[3,] -0.8210655 -0.1803023  0.2557724 -0.19263561  0.5525995 -0.4928963 1.1966553  0.40153119 -0.8013221
[4,] -0.3683845 -0.2523631 -0.8332908 -0.00501764 -0.6865140  0.6113654 0.7760149  0.60499691 -0.2004804
[5,] -0.5023781 -0.7297660  0.8003041 -0.83991762 -0.4627852  0.3518834 0.9672151 -1.27706093 -1.5589050
[6,] -0.7812296 -0.2793859  0.6641711 -1.12134458 -0.7897735  0.2562848 1.3496155 -0.05626666 -1.2454224
          [,10]      [,11]      [,12]      [,13]       [,14]       [,15]      [,16]       [,17]      [,18]
[1,]  2.6716856 -0.2249868 -0.9014919 -0.2539188 -0.04795369 -0.03500164 -0.4974748 -0.09125899 -0.2392593
[2,]  2.9854070 -0.2249868 -0.9014919 -0.2539188 -0.04795369 -0.03500164 -0.4974748 -0.09125899  4.1789269
[3,] -0.8432120 -0.2249868 -0.9014919 -0.2539188 -0.04795369 -0.03500164 -0.4974748 -0.09125899  4.1789269
[4,]  0.7984949 -0.2249868 -0.9014919 -0.2539188 -0.04795369 -0.03500164 -0.4974748 -0.09125899 -0.2392593
[5,]  0.1992262 -0.2249868 -0.9014919 -0.2539188 -0.04795369 -0.03500164  2.0098444 -0.09125899 -0.2392593
[6,]  0.1878043 -0.2249868 -0.9014919 -0.2539188 -0.04795369 -0.03500164  2.0098444 -0.09125899 -0.2392593
          [,19]      [,20]      [,21]       [,22]       [,23]       [,24]       [,25]      [,26]      [,27]
[1,] -0.2234819 -0.3093451 -0.2933388 -0.05676569 -0.06318727 -0.01749279 -0.01236832 -0.1563453 -0.1527135
[2,] -0.2234819 -0.3093451 -0.2933388 -0.05676569 -0.06318727 -0.01749279 -0.01236832 -0.1563453 -0.1527135
[3,] -0.2234819 -0.3093451 -0.2933388 -0.05676569 -0.06318727 -0.01749279 -0.01236832 -0.1563453 -0.1527135
[4,] -0.2234819 -0.3093451 -0.2933388 -0.05676569 -0.06318727 -0.01749279 -0.01236832 -0.1563453 -0.1527135
[5,] -0.2234819 -0.3093451 -0.2933388 -0.05676569 -0.06318727 -0.01749279 -0.01236832 -0.1563453 -0.1527135
[6,] -0.2234819 -0.3093451 -0.2933388 -0.05676569 -0.06318727 -0.01749279 -0.01236832 -0.1563453 -0.1527135
          [,28]      [,29]       [,30]      [,31]       [,32]       [,33]
[1,] -0.1349871 -0.1258904 -0.05254272 -0.1098832 -0.01749279 -0.04288122
[2,] -0.1349871 -0.1258904 -0.05254272 -0.1098832 -0.01749279 -0.04288122
[3,] -0.1349871 -0.1258904 -0.05254272 -0.1098832 -0.01749279 -0.04288122
[4,] -0.1349871 -0.1258904 -0.05254272 -0.1098832 -0.01749279 -0.04288122
[5,] -0.1349871 -0.1258904 -0.05254272 -0.1098832 -0.01749279 -0.04288122
[6,] -0.1349871 -0.1258904 -0.05254272 -0.1098832 -0.01749279 -0.04288122
```

```{r}
head(test_features)
```

```
             [,1]        [,2]        [,3]       [,4]        [,5]       [,6]       [,7]       [,8]       [,9]
[1,] -0.02072541  0.06290303 -0.15262629  0.5672172  0.07072202  0.4318741  0.7760149  0.9610619 -0.1221097
[2,] -1.06370256 -0.81083447 -0.01649339 -0.2676828 -0.47999511 -0.7621333  0.7377748 -0.6157974 -0.8535692
[3,] -0.14385466  1.27892944 -0.69715792 -1.0650592 -0.72093385  1.6161265 -0.6771067  0.7575962  1.0534501
[4,] -1.01662373 -0.78381166 -0.42489210 -0.7554895 -0.65209421 -0.7237638  0.6995348 -0.3105988 -0.6445807
[5,] -0.63637164 -0.88289529  1.48096858  0.1872908  1.00005716 -1.3162152  0.5083346 -2.2943895 -1.8462641
[6,] -0.84641565  1.39602828  1.75323439  0.3420756  2.06707158 -0.8070061 -2.8950291 -0.5649309  1.7849095
           [,10]      [,11]      [,12]      [,13]       [,14]       [,15]      [,16]       [,17]      [,18]
[1,] -0.28734646 -0.2249868 -0.9014919 -0.2539188 -0.04795369 -0.03500164 -0.4974748 -0.09125899 -0.2392593
[2,] -1.33816072 -0.2249868 -0.9014919 -0.2539188 -0.04795369 -0.03500164 -0.4974748 -0.09125899 -0.2392593
[3,] -0.07337631 -0.2249868 -0.9014919 -0.2539188 -0.04795369 -0.03500164 -0.4974748 -0.09125899 -0.2392593
[4,] -0.08099090 -0.2249868 -0.9014919 -0.2539188 -0.04795369 -0.03500164 -0.4974748 -0.09125899 -0.2392593
[5,] -0.04824814 -0.2249868 -0.9014919 -0.2539188 -0.04795369 -0.03500164  2.0098444 -0.09125899 -0.2392593
[6,] -0.93991735 -0.2249868  1.1091026 -0.2539188 -0.04795369 -0.03500164 -0.4974748 -0.09125899 -0.2392593
          [,19]      [,20]      [,21]       [,22]       [,23]       [,24]       [,25]      [,26]      [,27]
[1,] -0.2234819 -0.3093451 -0.2933388 -0.05676569 -0.06318727 -0.01749279 -0.01236832 -0.1563453 -0.1527135
[2,] -0.2234819 -0.3093451 -0.2933388 -0.05676569 -0.06318727 -0.01749279 -0.01236832 -0.1563453 -0.1527135
[3,] -0.2234819 -0.3093451 -0.2933388 -0.05676569 -0.06318727 -0.01749279 -0.01236832 -0.1563453 -0.1527135
[4,] -0.2234819 -0.3093451 -0.2933388 -0.05676569 -0.06318727 -0.01749279 -0.01236832 -0.1563453 -0.1527135
[5,] -0.2234819 -0.3093451 -0.2933388 -0.05676569 -0.06318727 -0.01749279 -0.01236832 -0.1563453 -0.1527135
[6,] -0.2234819 -0.3093451  3.4085061 -0.05676569 -0.06318727 -0.01749279 -0.01236832 -0.1563453 -0.1527135
          [,28]      [,29]       [,30]      [,31]       [,32]       [,33]
[1,] -0.1349871 -0.1258904 -0.05254272 -0.1098832 -0.01749279 -0.04288122
[2,] -0.1349871 -0.1258904 -0.05254272 -0.1098832 -0.01749279 -0.04288122
[3,] -0.1349871 -0.1258904 -0.05254272 -0.1098832 -0.01749279 -0.04288122
[4,] -0.1349871 -0.1258904 -0.05254272 -0.1098832 -0.01749279 -0.04288122
[5,] -0.1349871 -0.1258904 -0.05254272 -0.1098832 -0.01749279 -0.04288122
[6,] -0.1349871 -0.1258904 -0.05254272 -0.1098832 -0.01749279 -0.04288122
```

2) What is the rank of the tensor ‘training_features’? What is the shape of ‘training_features’? How many dimensions does ‘training_features’ have along the second axis?

Rank of Tensor: 2
Shape of training_feature: (6537, 33)
Dimesions of second axis of training_feature: 33

3) State two situations where scaling the numerical variables is important.

Situation one is when we are comparing models. When we scale our variables, when the variables are all in the same units (0 and standard deviation) we are able create consistency for two or more comparing mode.

Situation two is when we want to create algorithms such as k-mean clustering or k-nearest neighbours which are distance based algorithms. Without scaling, large scales can influence the distance calculations by masking important features by features with larger numerical ranges. But ensuring range of -1 to 1, this migtigates the issue.